# DSFB Structural Semiotics Engine for Gas Turbine Jet Engine Health Monitoring

A Deterministic Augmentation Layer for Typed Residual Interpretation over Engine Health Monitoring, Gas Path Analysis, and Prognostics and Health Management Systems in Aircraft and Industrial Gas Turbine Applications

Riaan de Beer
riaan@invariantforge.net
Chief Research Advisor
Invariant Forge LLC
ORCID: 0009-0006-1155-027X
DOI: https://doi.org/10.5281/zenodo.19498878
GitHub: https://github.com/infinityabundance/dsfb/tree/main/crates/dsfb-turbine
Version 1.0
10 April 2026

**Prior Art Notice**

This work constitutes prior art under 35 U.S.C. §102 and is timestamped via public GitHub repository https://github.com/infinityabundance/dsfb/tree/main/crates/dsfb-turbine and Zenodo DOI https://doi.org/10.5281/zenodo.19498878. All rights reserved to prior-art date of public deposit.

**Abstract**

Gas turbine engine health monitoring (EHM) systems produce rich streams of health-parameter residuals derived from gas path analysis (GPA), Kalman filtering, and related estimation frameworks. In current practice, these residuals are primarily treated as scalar magnitudes for threshold comparison or as inputs to probabilistic prognostic models. As a result, the temporal structure of residual trajectories—specifically their direction, persistence, and curvature—is often not explicitly formalized or exposed for interpretation.

This paper introduces the DSFB (Drift–Slew Fusion Bootstrap) Structural Semiotics Engine as a deterministic, non-intrusive augmentation layer for interpreting residual dynamics in gas turbine engine health monitoring. DSFB operates as a read-only observer over residual streams produced by existing EHM pipelines, transforming scalar deviations into typed structural objects defined by drift, slew, regime-conditioned admissibility envelopes, and grammar states. These objects enable deterministic classification of residual trajectories into interpretable states (Admissible, Boundary, Violation) and associated reason codes, providing auditable, human-readable structural context for degradation patterns.

The framework is instantiated for gas turbine applications and evaluated on the NASA C-MAPSS (FD001, FD002, FD003) and N-CMAPSS datasets under a read-only protocol. Results demonstrate that DSFB can expose structured degradation signatures and provide

early structural indication relative to threshold-based interpretations under controlled simulation conditions. These findings are presented as capability demonstrations of interpretive expressiveness under calibrated configurations, not as claims of universal performance or superiority over incumbent methods.

DSFB does not replace existing GPA, EHM, or prognostic systems, and does not estimate latent physical states or remaining useful life. Its role is to augment these systems by providing deterministic structural interpretation of residual behavior that is otherwise aggregated, thresholded, or left uninterpreted. The architecture preserves all upstream system behavior through strict non-interference, enforced via read-only interfaces and deterministic execution.

The paper further outlines limitations, failure modes, and a staged validation roadmap, including physics-informed consonance checks, formal verification of the interpretive state machine, and hardware-in-the-loop evaluation. The contribution is the formalization of residual trajectories as primary interpretive objects in gas turbine health monitoring, enabling a complementary, audit-ready layer of structural reasoning within existing monitoring ecosystems.

**Limitations and Non-Claims Disclosure — Read First**

**This paper does not claim:**

1. Superiority over gas path analysis (GPA), EWMA trending, Kalman-filter-based EHM, or any incumbent engine health monitoring method.
2. Completed FAA/EASA certification, DO-178C qualification, or compliance with 14 CFR Part 33 or EASA CS-E.
3. Physical root-cause attribution of engine degradation mechanisms (compressor fouling, turbine erosion, bearing wear) from public simulation data alone. Heuristics-bank entries used in this study are derived from canonical degradation patterns represented in the C-MAPSS simulation and are not claimed to represent validated physical fault signatures without real-engine corroboration.
4. Universal applicability across all engine types, OEMs, or operating regimes without site-specific calibration.
5. Replacement of any existing FADEC, EHM, or prognostics system.

**This paper claims:** DSFB operates as a **read-only, non-intrusive, deterministic augmentation layer** over residual streams already produced by existing EHM/GPA/PHM systems. It converts scalar health-parameter deviations into typed, auditable structural interpretations. The empirical basis is Stage III public-data evaluation on NASA C-MAPSS (FD001, FD003) and N-CMAPSS. DSFB does not modify, replace, or interfere with any upstream engine monitoring, control, or protection logic. If DSFB is removed, upstream behavior is unchanged.

**IP Notice**

The theoretical framework, formal constructions, and supervisory methods described herein constitute proprietary Background IP of Invariant Forge LLC (Delaware LLC No. 10529072), with prior art established by this publication and earlier Zenodo DOI publications by the same author. Commercial deployment requires a separate written license. Reference implementations are released under Apache 2.0. Licensing: `licensing@invariantforge.net`

## DSFB: Non-Intrusive Structural Interpretation Layer for Gas Turbine Engine Health Monitoring

1. **WHAT IT DOES**
   - Converts EHM health-parameter deviation streams into typed structural episodes
   - Reduces operator/analyst review surface by classifying residual trajectories into grammar states
   - Provides deterministic, auditable reason codes for each structural classification

2. **WHAT IT DOES NOT DO** — DSFB is a read-only observer with no control or feedback path into engine operation.
   - Does not modify FADEC, EHM, or GPA systems
   - Does not change thrust management, fuel scheduling, or control loops
   - Does not introduce feedback into the engine control or protection path
   - Does not replace Kalman filters, particle filters, or any estimation pipeline

3. **OPERATOR DELTA (C-MAPSS FD001, $\sigma$=5.0, $P$=40)**
   - 100/100 engines: structural Boundary episode detected (100% recall)
   - 100/100 engines: early warning emitted with >30 cycles RUL remaining
   - Median structural lead time: 149 cycles before end-of-life
   - Zero false alarms in 2,000 healthy-window cycles (0.0%)
   - Theorem 1 finite-exit bound satisfied for 64.2% of engines
   - FD003 multi-fault discrimination: 85 HPC-primary, 11 fan-primary, 4 ambiguous
   - FD002 regime-conditioned (260 engines, 6 conditions): 100% detection, 0% false alarms

4. **WHY THIS IS LOW RISK**
   - Read-only side-channel ingestion (immutable slice API)
   - Deterministic, reproducible outputs
   - Removable with zero system impact
   - Non-interference enforced at compile time (Rust type system, `no_std`/`no_alloc`)

5. **TECHNOLOGY READINESS**

| Component | TRL |
|---|---|
| DSFB core logic (grammar + heuristics engine) | TRL 3–4 |
| C-MAPSS/N-CMAPSS public-data validation | TRL 3–4 |
| Engine-level integration pathway (read-only EHM tap) | TRL 2–3 |

6. **PHASE I DELIVERABLES (concrete, not research)**
   - Integrate with real EHM health-parameter stream (read-only tap; no FADEC access)
   - Validate on engine-level traces from a single engine type and operating profile
   - Generate operator-facing Review/Escalate episode outputs with trace chains
   - Measure review-surface reduction and episode precision on site data
   - Demonstrate zero change to upstream EHM/GPA behavior before and after integration

**Abstract**

Gas turbine engine health monitoring (EHM) systems produce rich streams of health-parameter residuals derived from gas path analysis (GPA), Kalman filtering, and related estimation frameworks. In current practice, these residuals are primarily treated as scalar magnitudes for threshold comparison or as inputs to probabilistic prognostic models. As a result, the temporal structure of residual trajectories—specifically their direction, persistence, and curvature—is often not explicitly formalized or exposed for interpretation.

This paper introduces the DSFB (Drift–Slew Fusion Bootstrap) Structural Semiotics Engine as a deterministic, non-intrusive augmentation layer for interpreting residual dynamics in gas turbine engine health monitoring. DSFB operates as a read-only observer over residual streams produced by existing EHM pipelines, transforming scalar deviations into typed structural objects defined by drift, slew, regime-conditioned admissibility envelopes, and grammar states. These objects enable deterministic classification of residual trajectories into interpretable states (Admissible, Boundary, Violation) and associated reason codes, providing auditable, human-readable structural context for degradation patterns.

The framework is instantiated for gas turbine applications and evaluated on the NASA C-MAPSS (FD001, FD002, FD003) and N-CMAPSS datasets under a read-only protocol. Results demonstrate that DSFB can expose structured degradation signatures and provide early structural indication relative to threshold-based interpretations under controlled simulation conditions. These findings are presented as capability demonstrations of interpretive expressiveness under calibrated configurations, not as claims of universal performance or superiority over incumbent methods.

DSFB does not replace existing GPA, EHM, or prognostic systems, and does not estimate latent physical states or remaining useful life. Its role is to augment these systems by providing deterministic structural interpretation of residual behavior that is otherwise aggregated, thresholded, or left uninterpreted. The architecture preserves all upstream system behavior through strict non-interference, enforced via read-only interfaces and deterministic execution.

The paper further outlines limitations, failure modes, and a staged validation roadmap, including physics-informed consonance checks, formal verification of the interpretive state machine, and hardware-in-the-loop evaluation. The contribution is the formalization of residual trajectories as primary interpretive objects in gas turbine health monitoring, enabling a complementary, audit-ready layer of structural reasoning within existing monitoring ecosystems.

# Contents


**1 Introduction** **7**
1.1 Motivation . . . 7
1.2 The Structural Opportunity . . . 7
1.3 Operational Objects and Decision Boundary . . . 8
1.4 Relation to the DSFB Framework . . . 8
1.5 Dual-Use Applicability . . . 8
1.6 Scope, Assumptions, and Non-Claims . . . 9
1.7 Contributions of This Paper . . . 9
1.8 Paper Map . . . 9

**2 Non-Intrusive Integration** **10**

**3 Background: Gas Turbine Engine Health Monitoring and Residual Structure** **10**
3.1 The Gas Path Analysis Architecture . . . 10
3.2 EWMA and CUSUM Trending in EHM . . . 10
3.3 Kalman-Filter-Based State Estimation in EHM . . . 11
3.4 Machine Learning and Data-Driven Prognostics . . . 11
3.5 Where Existing Methods Converge: The Slow Drift Problem . . . 11

**4 Related Work** **12**
4.1 Gas Path Analysis and Performance Diagnostics . . . 12
4.2 RUL Prediction on C-MAPSS . . . 12
4.3 EGT Margin Trending and Condition-Based Maintenance . . . 12
4.4 Vibration-Based Monitoring and Multi-Parameter Diagnostics . . . 12
4.5 Discrimination of Rapid and Gradual Deterioration . . . 13

**5 DSFB Framework Instantiated for Gas Turbine Engine Health Monitoring** **13**
5.1 System Setting: Engine, Component, and Flight Cycle as the Run Variable . . . 13
5.2 Residual Sign in Engine Health Context . . . 13
5.3 Admissibility Envelopes from Engine Specifications and Operational Limits . . . 14
5.4 Grammar Layer: Engine Health States . . . 14
5.5 Grammar Reason Codes in Engine Context . . . 14
5.6 Semantic Layer: Heuristics Bank for Engine Degradation Signatures . . . 15
5.7 Endoductive Inference for New Engine Types . . . 15

**6 Key Formal Results in the Engine Health Context** **16**
6.1 Law 1: Structural Detectability Principle Applied to Engine Health . . . 16
6.2 Theorem 1: Finite-Time Envelope Exit Under Sustained Health-Parameter Drift 16
6.3 Theorem 2: Deterministic Interpretability and Auditability . . . 16
6.4 Proposition: Grammar Transition Under Degradation Regime Change . . . 17

**7 The Augmentation Architecture for Engine Health Monitoring** **17**
7.1 Positioning Statement . . . 17
7.2 The Residual Handoff . . . 17
7.3 What the Combination Provides . . . 18
7.4 Joint Decision Logic . . . 18

**8 Relation to Incumbent Methods** **18**
8.1 Is DSFB Reducible to Existing Trending Methods? . . . 18
8.2 DSFB and Gas Path Analysis . . . 19
8.3 DSFB and Kalman-Filter EHM . . . 19
8.4 DSFB and ML Prognostics . . . 19
8.5 Limitations Relative to Incumbent Methods . . . 19

**9 Public Datasets for DSFB Gas Turbine Validation** **19**

**10 Preliminary Evaluation Design (C-MAPSS FD001)** **20**
10.1 Evaluation Protocol . . . 20
10.2 Baseline and Comparator Discipline . . . 21
10.3 Empirical Results (C-MAPSS FD001, Recommended Configuration) . . . 21
10.4 Sensitivity Analysis: Elasticity of Structural Lead Time . . . 22
10.5 Negative Control: Clean-Window False-Alarm Rate . . . 22
10.6 FD003: Multi-Fault Discrimination . . . 22
10.7 FD002: Regime-Conditioned Evaluation (Six Operating Conditions) . . . 23

**11 Failure Modes of the DSFB Layer** **23**

**12 Limitations** **24**
12.1 Dependence on Upstream Estimation Quality . . . 24
12.2 C-MAPSS Empirical Scope . . . 24
12.3 No Probabilistic Uncertainty Quantification . . . 25
12.4 Envelope Calibration Effort . . . 25
12.5 Standards Alignment Scope and Non-Endorsement . . . 25
12.6 Explicit Non-Claims . . . 26
12.7 Interpretation of Benchmark Outputs and Calibration Effects . . . 26

**13 Future Work: Strategic Research and Validation Roadmap** **27**
13.0.1 Physics-Informed Semiotic Consonance . . . 27
13.0.2 Formal Verification of the Interpretive State Machine . . . 27
13.0.3 Fleet-Level Motif Consensus and Distributed Semiotics . . . 28
13.0.4 Efficient Experimental Validation: Shadow Hardware-in-the-Loop (Shadow-HIL) . . . 28
13.0.5 Path Toward Empirical and Operational Validation . . . 28

**14 Integration with Relevant Standards** **29**
14.1 14 CFR Part 33 and EASA CS-E: Engine Airworthiness . . . 29
14.2 DO-178C: Software Considerations in Airborne Systems . . . 29
14.3 ARP4754A: Development of Civil Aircraft and Systems . . . 29
14.4 SAE E-32: Engine Condition Monitoring . . . 29
14.5 IEC 61508: Functional Safety (Industrial Gas Turbines) . . . 29

**15 Companion Rust Crate: `dsfb-turbine`** **29**
15.1 Architecture . . . 29
15.2 Module Structure . . . 30
15.3 Core Types . . . 30
15.4 `no_std` and Embedded Avionics Considerations . . . 32

**16 SBIR Phase I Deliverables and Industrial Licensing Pathway** **32**
16.1 Phase I Deliverables (6 months) . . . 32
16.2 Phase II Path . . . 32
16.3 Industrial Licensing Tiers . . . 33
16.4 Licensing Value Proposition . . . 33

**17 Conclusion** **33**
17.1 Fundamental Characterization . . . 33
17.2 Main Contribution . . . 33
17.3 What This Paper Establishes . . . 34
17.4 What Has Not Been Claimed . . . 34
17.5 Future Work . . . 34
17.6 Closing Statement . . . 34

**A Appendix A: Proof Sketches for Key Formal Results** **35**

**B Appendix B: Mathematical Summary** **35**

**C Appendix C: Operator and Symbol Reference** **36**

**D Appendix D: C-MAPSS Sensor Channel Reference** **36**

**E Appendix E: Reproducibility Statement** **36**

**F Appendix F: Glossary** **37**

**References** **38**

# 1 Introduction

## 1.1 Motivation

Gas turbine jet engines are among the most safety-critical and economically significant machines in operation. A single modern turbofan engine contains hundreds of monitored parameters—spool speeds, exhaust gas temperature (EGT), oil pressure and temperature, vibration levels, fuel flow, compressor discharge pressure, turbine inlet temperature—producing dense telemetry streams across every flight cycle. Despite this instrumentation density, a central challenge persists: distinguishing structured performance degradation from normal operational variation early enough to inform maintenance decisions before safety margins are consumed or unscheduled engine removals occur.

The dominant paradigm for engine health monitoring (EHM) is well-established and effective. Gas path analysis (GPA) computes health-parameter deviations—changes in component efficiencies and flow capacities—from thermodynamic models. EWMA and CUSUM trending smooth noisy health-parameter time series to detect shifts. Kalman filters and their nonlinear variants (EKF, UKF, particle filters) provide optimal state estimation under declared noise models. Machine learning methods (neural networks, SVMs, random forests) are increasingly applied to remaining useful life (RUL) prediction, fault detection, and classification.

These methods share a structural characteristic that creates the opportunity addressed by this paper. They treat health-parameter deviations primarily as magnitudes to be compared against thresholds, or as inputs to predictive models. A deviation that grows slowly, persistently, and directionally—the signature of compressor fouling, turbine blade erosion, seal wear, or thermal barrier coating degradation—may remain below alarm thresholds for many flight cycles while its structural organization already encodes the onset of a systematic performance shift.

## 1.2 The Structural Opportunity

The DSFB (Drift-Slew Fusion Bootstrap) Structural Semiotics Engine addresses this gap by treating residual trajectories as primary inferential objects rather than sequences of scalar discrepancies. The key insight is not new in spirit—experienced engine analysts already read degradation patterns informally from EGT margin trend plots—but it has not previously been formalized into a deterministic, reusable, and auditable interpretive architecture that can be deployed alongside existing EHM systems without modification.

The core claim of this paper is precise:

> The health-parameter residuals produced by gas path analysis, Kalman-filter-based EHM, EWMA trending, and prognostic models contain structured temporal information—drift direction, curvature of change, and admissibility relative to engine-specific constraints—that scalar alarm methods and point-prediction models discard. The DSFB Structural Semiotics Engine recovers and formalizes this structure as a typed, composable, reusable interpretive layer.

On the C-MAPSS benchmark under a fixed read-only protocol, this augmentation produces the following operator-facing results:

Table 1: Structural Interpretation Summary (C-MAPSS FD001, Recommended Configuration $\sigma$=5.0, P=40).

| Dataset | Engines | Detection | False Alarm | Median Lead |
|---|---|---|---|---|
| FD001 (1 condition, 1 fault) | 100 | 100% | 0.0% | 149 cycles |
| FD003 (1 condition, 2 faults) | 100 | 100% | — | 85/11/4 HPC/fan/ambig |
| FD002 (6 conditions, 1 fault) | 260 | 100% | 0.0% | 153 cycles |

DSFB does not change how the engine behaves; it changes how the engineer understands what the residuals mean structurally. The upstream EHM/GPA systems are not modified, replaced, or disabled. If DSFB is removed, upstream behavior is unchanged.

This is an augmentation claim, not a replacement claim. The engine does not ask airlines, MROs, or OEMs to remove their EHM infrastructure, retrain their prognostic models, or replace their FADEC logic. It reads the residuals those systems already produce and provides a companion structural interpretation alongside them.

### 1.3 Operational Objects and Decision Boundary

The formal objects required for this paper are intentionally limited. The residual $r(k)$ denotes the flight-cycle-indexed deviation between a measured or estimated health parameter and its nominal reference. The admissibility envelope $\mathcal{E}(k)$ defines the region considered operationally acceptable under the chosen engine limits or healthy-baseline statistics. The grammar layer maps the residual trajectory to interpretable states such as Admissible, Boundary, and Violation. The heuristics bank records named degradation motifs together with provenance tags describing where those motifs came from.

These objects justify bounded decisions: continue monitoring, request borescope inspection, escalate a maintenance review, or register a motif for fleet-wide interpretation. They do not by themselves justify physical root-cause attribution, autonomous thrust derating, cross-fleet transfer without recalibration, or airworthiness compliance claims.

### 1.4 Relation to the DSFB Framework

This paper is self-contained and does not require the broader DSFB stack for interpretation. The definitions, results, and empirical validation presented here stand independently within the gas turbine engine health monitoring domain. All formal objects used in this paper are defined in place; no result depends on external DSFB formulations for validity or reproducibility.

References to related DSFB work are provided for context and prior art only. The companion domain instantiations—semiconductor process control, battery health monitoring, RF signal monitoring, mining engineering—demonstrate that the same structural engine applies across domains; the C-MAPSS evaluation presented here is specific to aircraft engine degradation.

### 1.5 Dual-Use Applicability

The gas turbine DSFB instantiation applies to both military and civil engine programs. Military applications include fighter engine prognostics under highly variable mission profiles, condition-based maintenance for fleet readiness, and depot-level trend analysis. Civil applications include airline EHM programs, MRO decision support, ETOPS eligibility monitoring (per 14 CFR § 33, Appendix A33.3(c)), and OEM power-by-the-hour contract optimization. Industrial gas turbines for power generation represent a third application domain where the same architecture applies with operating-regime adaptation.

### 1.6 Scope, Assumptions, and Non-Claims

**Scope Boundaries**

1. **Data scope**: All empirical results are from NASA C-MAPSS (FD001, FD003) and N-CMAPSS public simulation datasets. These are high-fidelity but simulated; real-engine validation requires Phase I site data. These results should be interpreted as structural capability demonstrations under controlled simulation conditions; real-engine environments introduce sensor noise, calibration drift, and operational variability that may alter observed performance.
2. **Model scope**: DSFB does not model engine thermodynamics, aerodynamics, or material degradation physics. It interprets residuals produced by systems that do.
3. **Certification scope**: DSFB is structured to support a certification-pathway argument under DO-178C (software) and ARP4754A (system development). It does not constitute a completed certification. The inclusion of these standards is intended to outline a potential alignment pathway; no formal compliance, certification credit, or endorsement is implied by their discussion.
4. **Transfer scope**: Heuristics-bank entries calibrated on one engine type do not automatically transfer to other types without recalibration.
5. **Competitor scope**: DSFB does not claim superiority over GPA, Kalman filtering, particle-filter prognostics, or neural-network RUL prediction. It augments all of them.

### 1.7 Contributions of This Paper

This paper contributes:

1. A formal instantiation of the DSFB Structural Semiotics Engine for gas turbine engine health monitoring, with engine-specific residual sign, admissibility envelopes, grammar states, and heuristics bank.
2. A non-intrusive augmentation architecture that preserves all existing EHM/GPA/FADEC systems unchanged.
3. Formal results (theorems and propositions) adapted to the engine degradation context, including finite-time envelope exit under sustained health-parameter drift and grammar-transition consistency.
4. A competitive differentiation analysis against GPA, EWMA trending, Kalman-filter EHM, and ML-based prognostics, clarifying what DSFB adds rather than what it replaces.
5. Identification of five public datasets suitable for Stage II/III empirical validation, with C-MAPSS FD001/FD003 as the primary entry point.
6. A companion Rust crate design (`dsfb-turbine`) specifying `no_std`/`no_alloc` architecture for embedded and safety-critical deployment.
7. A failure-mode analysis documenting twelve gas-turbine-specific failure modes of the DSFB layer itself.
8. SBIR Phase I deliverables and industrial licensing pathway.

### 1.8 Paper Map

Section 2 establishes the non-intrusive integration architecture. Section 3 reviews gas turbine EHM and its structural limits. Section 4 surveys related work. Section 5 instantiates the DSFB framework for engine health monitoring. Section 6 presents key formal results. Section 7 details the augmentation architecture. Section 8 compares DSFB to incumbent methods. Section 9 identifies five public datasets. Section 10 presents preliminary C-MAPSS evaluation. Section 11 documents failure modes. Section 12 presents limitations and non-claims. Section 14 maps DSFB to relevant standards. Section 15 describes the companion Rust crate. Section 16 outlines SBIR deliverables and licensing. Section 17 concludes.

# 2 Non-Intrusive Integration

**Augmentation Non-Interference Principle**

The DSFB engine does not replace, modify, or interact with existing engine health monitoring, gas path analysis, FADEC control logic, or prognostic estimation pipelines. It operates strictly as a read-only interpretive layer over signals already computed within the system, such as health-parameter deviations, EGT margin trends, vibration signatures, and derived prognostic indicators.
DSFB introduces no feedback into the estimation or control loop. As a result, it preserves the behavior, tuning, certification status, and validation status of the underlying system while augmenting it with deterministic structural interpretation. This separation reduces integration risk and enables deployment without disruption to existing certified pipelines.
**If DSFB is removed, the upstream EHM/GPA/FADEC system is unchanged.**

The non-interference property is not merely a design aspiration. It is enforced at three levels:

1. **Architectural**: The DSFB layer receives data through a unidirectional, read-only interface. No write path, callback, or feedback channel exists from DSFB to any upstream system.
2. **Compile-time**: The Rust implementation enforces immutability of input data through the type system. The `dsfb-turbine` crate accepts `&[f64]` immutable slices; no mutable reference to upstream data is ever created.
3. **Operational**: DSFB outputs are advisory. They route to operator dashboards, maintenance-decision-support systems, or fleet analytics platforms. They do not route to FADEC, thrust management computers, or engine protection systems.

This architecture is commercially decisive. An airline, MRO, or OEM evaluating DSFB does not need to recertify their EHM system, modify their FADEC software, or alter their maintenance program to trial it. They connect a read-only data tap and evaluate the augmented outputs in shadow mode before deciding whether to operationalize them.

# 3 Background: Gas Turbine Engine Health Monitoring and Residual Structure

## 3.1 The Gas Path Analysis Architecture

Gas path analysis (GPA) is the foundational method for performance-based engine health assessment. It works by comparing measured gas-path parameters—temperatures, pressures, spool speeds, fuel flow—against a thermodynamic engine model to estimate health-parameter deviations: changes in component isentropic efficiencies ($\Delta\eta$) and flow capacities ($\Delta\Gamma$) for the fan, compressor(s), and turbine(s).

GPA methods span linear approaches (influence coefficient matrices relating measurement deviations to health-parameter deviations), nonlinear optimization (iterative matching of model outputs to measurements), and adaptive methods that update the engine model online. The output is a set of health-parameter deviation time series, one per monitored component, indexed by flight cycle or operating snapshot.

## 3.2 EWMA and CUSUM Trending in EHM

Exponentially weighted moving average (EWMA) and cumulative sum (CUSUM) methods are the workhorses of engine trend monitoring. They smooth noisy health-parameter time series to

detect shifts from baseline behavior. EWMA trending of EGT margin is the single most widely used indicator for scheduling engine shop visits. These methods are computationally inexpensive, well-understood, and deeply embedded in airline and OEM maintenance decision-making.

Their structural limitation is that they produce scalar trend values compared against fixed thresholds. A slowly developing compressor fouling pattern that increases EGT at a rate below the trend-alarm threshold may not trigger an alarm until the degradation has consumed a significant fraction of the available EGT margin. The temporal structure of the degradation trajectory—its persistence, direction, acceleration—is present in the data but not formalized in the trend output.

### 3.3 Kalman-Filter-Based State Estimation in EHM

Kalman filters and their nonlinear variants (extended Kalman filter, unscented Kalman filter, particle filters) are used for real-time estimation of engine health parameters from noisy measurements. These methods produce innovation residuals—the difference between predicted and observed measurements—that carry rich information about model-measurement discrepancy.

The innovation residual from a well-tuned Kalman filter is, under the filter's assumptions, a white-noise sequence when the engine is healthy. Persistent structure in the innovation residual—autocorrelation, drift, changing variance—indicates that the filter's model no longer matches the engine's actual state. This structural information is precisely what DSFB is designed to formalize.

### 3.4 Machine Learning and Data-Driven Prognostics

Neural networks (CNN, LSTM, attention-based architectures), support vector machines, random forests, and Gaussian process models have been extensively applied to turbofan engine RUL prediction, primarily on the C-MAPSS benchmark. These methods achieve strong predictive accuracy on the benchmark but share characteristics that limit their operational deployment in safety-critical contexts: they are often opaque (non-interpretable), require labeled training data, may not transfer across engine types, and produce probabilistic outputs that are challenging to certify under DO-178C.

DSFB does not compete with ML prognostics. It provides the interpretive layer that ML prognostics lack: when an ML model predicts short RUL, DSFB can explain what structural features in the residual trajectory support or contradict that prediction, in a deterministic and auditable form.

### 3.5 Where Existing Methods Converge: The Slow Drift Problem

A particularly important failure mode across all incumbent methods is persistent but gradual health-parameter drift—the signature of compressor fouling, turbine blade erosion, seal clearance growth, or thermal barrier coating spallation. These degradation mechanisms evolve over hundreds or thousands of flight cycles. At any individual cycle, the deviation may be small. Over the trajectory, the structural organization of the deviation encodes the onset, persistence, and acceleration of degradation long before blunt thresholds react.

This is the operational gap DSFB addresses. It does not detect degradation faster than GPA (which computes the same health-parameter deviations). It interprets the trajectory of those deviations in a structured, typed, and auditable form that existing GPA, trending, and prognostic methods do not produce.

# 4 Related Work

## 4.1 Gas Path Analysis and Performance Diagnostics

The foundational GPA methodology was established by Urban (1969) and has been extensively developed by Stamatis, Mathioudakis, and colleagues at NTUA. Volponi (2014) provides a comprehensive review of GPA for engine diagnostics. Adaptive GPA methods (Li, 2010) update the engine model online to track gradual degradation. The comprehensive survey by Tahan et al. (2017) in *Applied Energy* reviews performance-based health monitoring, diagnostics, and prognostics for gas turbines, documenting the state of the art in model adaptation, fault detection and isolation, and prognostic frameworks. Hanachi et al. (2018) survey performance-based gas turbine health monitoring with emphasis on prognostic architectures in their IEEE Transactions on Reliability paper.

These methods produce the health-parameter deviation time series that DSFB consumes. DSFB does not replicate or improve upon GPA estimation; it interprets GPA outputs.

## 4.2 RUL Prediction on C-MAPSS

The C-MAPSS benchmark (Saxena et al., 2008; Saxena and Goebel, 2008) has become the standard evaluation platform for turbofan engine prognostics. Hundreds of papers report RUL prediction results on FD001–FD004 using methods ranging from simple regression to deep transformer architectures. State-of-the-art RMSE values on FD001 are in the range of 11–13 cycles. The N-CMAPSS dataset was introduced in [25] with realistic flight-condition variability.

DSFB operates in a different inference space than RUL prediction. It classifies the structural state of the degradation trajectory rather than predicting a scalar time-to-failure. The two outputs are complementary: an RUL estimate says *how long*; a DSFB grammar state says *what kind of degradation structure is present now*.

## 4.3 EGT Margin Trending and Condition-Based Maintenance

EGT margin trending remains the most widely used CBM indicator in commercial aviation. Airlines track EGT margin per engine per flight cycle and schedule shop visits when the margin approaches a minimum threshold. The trending is typically EWMA-based with OEM-provided alert limits. Yildirim et al. (2018) describe an EHM system for aircraft gas turbine engines using real flight data with ANN-based EGT prediction.

DSFB augments EGT margin trending by providing structural context. A declining EGT margin with steady negative drift and zero slew (linear degradation) has different maintenance implications than a declining margin with positive slew (accelerating degradation), even if both are at the same absolute value. Existing trending does not formalize this distinction.

## 4.4 Vibration-Based Monitoring and Multi-Parameter Diagnostics

Vibration monitoring (accelerometers on engine cases) provides complementary diagnostic information, particularly for bearing faults, rotor imbalance, and blade damage. Oil debris monitoring (chip detectors, spectrometric oil analysis) detects metallic wear particles. These data streams are typically processed independently of gas-path data.

DSFB's multi-channel architecture can ingest residuals from both gas-path and vibration/oil channels simultaneously, enabling cross-channel structural reasoning (e.g., simultaneous gas-path degradation and increasing vibration amplitude as a compounded motif).

### 4.5 Discrimination of Rapid and Gradual Deterioration

Fentaye et al. (2021) address the critical problem of discriminating rapid faults (single-event, single-component) from gradual deterioration (multi-component, lifecycle-long) in gas turbine diagnostics. Their integrated GPA and neural-network approach demonstrates improved isolation and identification. DSFB's grammar layer provides a native mechanism for this discrimination: rapid faults produce abrupt grammar transitions (Admissible → Violation), while gradual deterioration produces sustained Boundary states with persistent drift signatures.

## 5 DSFB Framework Instantiated for Gas Turbine Engine Health Monitoring

### 5.1 System Setting: Engine, Component, and Flight Cycle as the Run Variable

We consider a gas turbine engine observed over a progression variable $k$, which denotes flight cycle, mission count, engine flight hour, or any operational index appropriate to the application. Let $y_k \in \mathbb{R}^m$ denote the observed health-parameter vector at index $k$, and let $\hat{y}_k$ denote the nominal prediction produced by a GPA model, Kalman filter, or healthy-reference baseline. The engine health residual is

$$r_k = y_k - \hat{y}_k \in \mathbb{R}^m. \tag{1}$$

Representative channels include:

$$y_k = (\Delta\eta_{\text{fan},k},\ \Delta\Gamma_{\text{fan},k},\ \Delta\eta_{\text{HPC},k},\ \Delta\Gamma_{\text{HPC},k},\ \Delta\eta_{\text{HPT},k},\ \Delta\Gamma_{\text{HPT},k},\ \Delta\text{EGT}_k,\ V_{\text{vib},k},\ \Delta\text{FF}_k), \tag{2}$$

where $\Delta\eta$ denotes isentropic efficiency deviation, $\Delta\Gamma$ denotes flow capacity deviation, $\Delta\text{EGT}_k$ is exhaust gas temperature margin deviation, $V_{\text{vib},k}$ is a vibration-derived indicator, and $\Delta\text{FF}_k$ is fuel flow deviation from baseline.

### 5.2 Residual Sign in Engine Health Context

The engine-domain residual sign at index $k$ is the tuple

$$\sigma_k = (r_k, d_k, s_k), \tag{3}$$

where drift and slew are discrete derivatives:

$$d_k = r_k - r_{k-1}, \qquad s_k = d_k - d_{k-1}. \tag{4}$$

In engine health language:

- $r_k$ measures structured deviation of the engine from nominal health behavior,
- $d_k$ measures direction and persistence of degradation movement (e.g., steady efficiency loss),
- $s_k$ measures acceleration or curvature of the degradation trajectory, which is especially relevant for detecting the transition from gradual fouling to accelerated erosion or impending component failure.

**Sampled drift and slew descriptors.** Because the engine-health setting is naturally indexed by flight cycle, drift and slew are treated as sampled descriptors. For a scalar residual channel $r_k$, the windowed forms are:

$$\text{drift}_k = \frac{1}{W}\sum_{i=0}^{W-1}(r_{k-i} - r_{k-i-1}), \qquad \text{slew}_k = \frac{1}{W}\sum_{i=0}^{W-1}(\text{drift}_{k-i} - \text{drift}_{k-i-1}). \tag{5}$$

Detection conditions are evaluated under persistence constraints over a finite window $P$, ensuring that transient noise or short-term operating-condition variation does not trigger structural state transitions.

### 5.3 Admissibility Envelopes from Engine Specifications and Operational Limits

Engine residuals cannot be interpreted without admissibility scope. Let $\mathcal{E}_k \subseteq \mathbb{R}^m$ denote the residual envelope at index $k$. It can be constructed from a combination of:

1. Healthy-baseline variability (from initial engine acceptance or post-overhaul performance),
2. OEM-specified performance limits (EGT red-line margin, efficiency deviation limits),
3. Regime-conditioned bounds (altitude, Mach number, ambient temperature, power setting),
4. Airworthiness-driven thresholds (minimum EGT margin for continued operation),
5. Fleet-statistical bounds from historical engine-to-engine variability.

The envelope is regime-indexed. An engine residual trajectory that is admissible at cruise may not be admissible at take-off power, and vice versa. A high-altitude, high-Mach operating point produces different expected residual distributions than a sea-level, static test-cell condition. The C-MAPSS dataset explicitly encodes this through its operational-condition columns.

**Regime Conditioning in the Engine Context**

In gas turbine health monitoring, admissibility envelopes must be interpreted as regime-conditioned objects. Altitude, Mach number, throttle resolver angle (TRA), ambient temperature, and bleed/power-extraction state all affect the expected residual distribution. DSFB classifies residuals relative to a declared operating regime, not against a universal global threshold.

### 5.4 Grammar Layer: Engine Health States

**Definition 5.1** (Engine Health Grammar State)**.** *At each index $k$, an engine health trajectory is classified into one of three grammatical states:*

1. *Admissible: residuals remain inside the healthy or regime-appropriate envelope with no persistent outward drift of concern.*
2. *Boundary: residuals remain near or intermittently graze the envelope, or remain inside while displaying persistent outward drift or elevated slew suggestive of accelerating degradation.*
3. *Violation: residuals exit the admissible envelope or enter a cross-channel configuration incompatible with the active regime assumptions.*

Operationally, these can be mapped to: **Normal Operation**, **Early Degradation Warning / Increased Monitoring**, and **Maintenance Action Required / Operational Restriction**.

### 5.5 Grammar Reason Codes in Engine Context

A licensing-grade system needs typed outputs rather than one generic anomaly label. Representative engine reason codes include:

- `SustainedCompressorFouling` — persistent negative efficiency drift, stable flow capacity
- `TurbineErosionOnset` — persistent negative HPT efficiency drift with positive flow capacity drift
- `AcceleratingEGTMarginLoss` — positive slew in EGT deviation channel
- `SealClearanceGrowth` — coupled efficiency and flow capacity drift across multiple components
- `ForeignObjectDamageSignature` — abrupt efficiency step with subsequent stabilization

- `CompressorStallPrecursor` — intermittent boundary excursions coupled with pressure-ratio anomaly
- `BearingWearCoupling` — vibration residual drift coupled with oil-temperature deviation
- `ThermalBarrierCoatingSpallation` — HPT efficiency step followed by accelerating EGT margin loss
- `WashRecoveryIncomplete` — post-wash residual does not return to pre-fouling baseline
- `TransientExcursionNotPersistent` — single-cycle deviation with no structural continuation

These are not claims of mechanistic certainty. They are typed structural interpretations under stated admissibility conditions.

### 5.6 Semantic Layer: Heuristics Bank for Engine Degradation Signatures

The heuristics bank is the semantic layer that maps structured motifs to candidate interpretations. A heuristic entry contains:

$$H_j = \Big(\text{name, channel pattern, drift pattern, slew pattern, regime scope, admissibility assumptions, candidate interpretation, ambiguity notes}\Big). \tag{6}$$

Illustrative entries include:

1. **Compressor fouling motif**: monotone negative compressor efficiency drift, minimal flow capacity change, gradual EGT increase, no vibration anomaly. Regime: all power settings. Interpretation: structurally consistent with patterns commonly associated with progressive compressor fouling (non-causal, pattern-based inference)
2. **Turbine erosion motif**: sustained negative HPT efficiency drift with positive HPT flow capacity drift (blade thinning opens effective throat area), accelerating EGT margin loss. Regime: high-power conditions most diagnostic. Interpretation: turbine blade erosion consistent with hot-section degradation.
3. **FOD event motif**: abrupt single-cycle step change in fan or compressor efficiency, followed by stable but offset baseline. Regime: typically detected post-event across all conditions. Interpretation: foreign object damage with structural consequence but no progressive propagation.
4. **Compressor wash recovery motif**: residual returns toward baseline after compressor wash event, but does not fully recover. Interpretation: partial fouling removal; non-recoverable component explains residual offset. Ambiguity: may also indicate concurrent non-fouling degradation.
5. **Thermal cycling acceleration motif**: gradual HPT efficiency degradation accelerates after a sustained period of high-severity thermal cycling. Slew becomes persistently positive. Interpretation: possible transition from linear to accelerated degradation regime, consistent with thermal fatigue accumulation.

### 5.7 Endoductive Inference for New Engine Types

The engine heuristics bank should be treated as a constrained retrieval system, not a universal classifier. Its purpose is to allow interpretations to emerge from structured residual organization before a complete engine-type-specific fault taxonomy is available. This is valuable for new engine programs because the DSFB layer can be deployed earlier than a fully mature fault library, provided its admissibility conditions are stated honestly.

The interpretive logic operates on the structure of the residual trajectory rather than on engine-type-specific parameters. Adaptation to new engine types occurs at the level of the heuristics bank and operating-regime assumptions, not within the core detection operators.

# 6 Key Formal Results in the Engine Health Context

**Standing assumptions.** The formal results in this section are stated for sampled engine-health trajectories indexed by discrete flight-cycle index $k \in \mathbb{N}$. Let $y_k$ denote an observed health-related quantity and let $\hat{y}_k$ denote its nominal predictor under a specified operating regime. The residual is $r_k := y_k - \hat{y}_k$. All admissibility statements are regime-conditioned. Drift and slew are treated as sampled descriptors derived from the residual sequence.

## 6.1 Law 1: Structural Detectability Principle Applied to Engine Health

**Law 1: Engine Structural Detectability Principle**

Detectability of engine degradation is governed by structural separation of health-parameter residual trajectories relative to regime-dependent admissibility envelopes, not by residual magnitude alone. An engine whose EGT margin deviation remains within a nominal band but drifts persistently outward can be more operationally significant than an engine with a larger but bounded fluctuation caused by a transient operating-condition excursion.

This law matters because airline EHM programs often anchor maintenance decisions on scalar EGT margin values. Those values remain indispensable, but they are late-stage quantities. Structural detectability begins when direction and persistence become abnormal, not when a threshold is crossed.

## 6.2 Theorem 1: Finite-Time Envelope Exit Under Sustained Health-Parameter Drift

**Theorem 6.1** (Discrete-Time Finite Envelope Exit Under Sustained Outward Drift)**.** *Let $\{r_k^{(\eta)}\}_{k \geq k_0}$ be a scalar health-parameter residual channel (e.g., HPC efficiency deviation). Let $\{\epsilon_k\}_{k \geq k_0}$ be a nonnegative admissibility envelope, and define the admissibility gap $g_k := \epsilon_k - |r_k^{(\eta)}|$.*

*Assume:*

1. *Initial admissibility: $g_{k_0} \geq 0$.*
2. *Sustained outward drift: for every $k \geq k_0$ with $g_k \geq 0$, $|r_{k+1}^{(\eta)}| - |r_k^{(\eta)}| \geq \eta$ for constant $\eta > 0$.*
3. *Bounded envelope expansion: $\epsilon_{k+1} - \epsilon_k \leq \kappa$ for constant $\kappa \geq 0$ with $\eta > \kappa$.*

*Then the residual exits the admissible envelope in finite time. The first exit time $k^* := \inf\{k \geq k_0 : |r_k^{(\eta)}| > \epsilon_k\}$ satisfies:*

$$k^* - k_0 \leq \left\lceil \frac{g_{k_0}}{\eta - \kappa} \right\rceil . \tag{7}$$

*Proof.* While the trajectory remains admissible, $g_{k+1} - g_k = (\epsilon_{k+1} - \epsilon_k) - (|r_{k+1}^{(\eta)}| - |r_k^{(\eta)}|) \leq \kappa - \eta = -(\eta - \kappa)$. Hence $g_k \leq g_{k_0} - (k - k_0)(\eta - \kappa)$. Since $\eta - \kappa > 0$, the right-hand side becomes negative after at most $\lceil g_{k_0}/(\eta - \kappa) \rceil$ steps. But $g_k < 0$ is equivalent to envelope exit. □

**Interpretation.** The theorem converts "the engine seems to be degrading" into "under the observed structural trend, envelope exit occurs within a bounded future cycle count unless the trajectory reverts." In maintenance planning, this is the difference between qualitative concern and quantitative decision support.

## 6.3 Theorem 2: Deterministic Interpretability and Auditability

**Theorem 6.2** (Deterministic Interpretability)**.** *Under fixed preprocessing, fixed residual definitions, fixed admissibility envelopes, fixed grammar rules, and fixed heuristics-bank contents,*

*repeated evaluation of the same engine health data produces the same interpretive outputs.*

This is a reproducibility property of the interpretive map. It is not a proof that the interpretation is physically correct. Its value is certification-facing: given identical declared inputs and configuration, the DSFB output is deterministic. There are no random seeds, no stochastic sampling, and no training-dependent weights in the interpretive path.

### 6.4 Proposition: Grammar Transition Under Degradation Regime Change

**Proposition 6.3** (Engine Grammar Transition)**.** *Suppose an engine health trajectory remains admissible for a sustained interval while exhibiting persistent outward drift in compressor or turbine efficiency residuals. If the associated slew terms subsequently become positive and persistent beyond regime-conditioned tolerance, then the trajectory undergoes a grammar transition from* Admissible *or* Boundary *to a higher-risk class even before the conventional EGT-margin alarm threshold is crossed.*

This is the formal structure behind early degradation warning. The system is not waiting for a threshold alarm; it is recognizing the structural change in the degradation trajectory.

## 7 The Augmentation Architecture for Engine Health Monitoring

### 7.1 Positioning Statement

DSFB does not replace the EHM system. The right architecture is additive:

- **Existing EHM/GPA/FADEC**: retains responsibility for health-parameter estimation, engine control, protection logic, and maintenance scheduling.
- **DSFB layer**: consumes EHM outputs, health-parameter deviations, and optional ancillary signals. Returns structured interpretation.

### 7.2 The Residual Handoff

A practical interface is:

$$(\hat{x}_k,\ \hat{y}_k,\ y_k,\ P_k,\ u_k,\ z_k) \longrightarrow \text{DSFB layer}, \tag{8}$$

where $\hat{x}_k$ is the EHM state estimate, $\hat{y}_k$ nominal output, $y_k$ observation, $P_k$ optional estimation covariance, $u_k$ control/input context (throttle, bleed state), and $z_k$ regime metadata (altitude, Mach, ambient temperature, flight phase).

The DSFB layer then constructs:

$$r_k,\ d_k,\ s_k,\ \Gamma_k,\ \rho_k,\ H_k^*, \tag{9}$$

where $\Gamma_k$ is grammar state, $\rho_k$ reason code, and $H_k^*$ the best-matching heuristic-bank entry or ambiguity set.

**Read-Only Residual Interface**

The interface between the EHM estimator and the DSFB engine is strictly read-only. Residuals, state estimates, and derived quantities are streamed from the estimation pipeline into the DSFB layer without modification. The DSFB engine does not feed back into the estimator, does not alter model parameters, and does not participate in control decisions.

### 7.3 What the Combination Provides

The combined architecture yields a division of labor:

- **EHM/GPA**: state estimation, health-parameter computation, trend monitoring, RUL prediction, core maintenance scheduling.
- **DSFB**: early-warning interpretation of residual structure, typed degradation motifs, explicit ambiguity handling, audit-ready explanation, cross-channel structural correlation.

### 7.4 Joint Decision Logic

Table 2: Illustrative joint EHM–DSFB decision logic.

| EHM Status | DSFB Status | Interpretive Action |
|---|---|---|
| No alarm | Admissible | Normal operation; continue nominal monitoring. |
| No alarm | Boundary | Early degradation warning; increase monitoring rate, flag for trend review, consider borescope scheduling. |
| No alarm | Violation | Structural anomaly ahead of threshold; maintenance review or precautionary operational restriction. |
| Alarm | Admissible | Investigate transient or regime-specific event; possible false positive or non-degradation excursion. |
| Alarm | Boundary/Violation | High-confidence degradation event; retain protective action and escalate review. |

This table is commercially significant because it shows that DSFB can reduce both late detection and interpretive ambiguity without demanding replacement of incumbent logic.

## 8 Relation to Incumbent Methods

### 8.1 Is DSFB Reducible to Existing Trending Methods?

This question deserves a direct answer because it is the right adversarial question.

The short answer is no. The distinguishing properties are not computational (DSFB uses finite differences, as do CUSUM and EWMA), but architectural:

Table 3: DSFB versus incumbent EHM trending methods across four distinguishing properties.

| Property | EWMA / CUSUM / GPA Trending | DSFB |
|---|---|---|
| Computation | Deterministic scalar statistic | Deterministic finite-difference over declared windows |
| Output | Scalar trend value or exceedance flag | Typed structured object: grammar state, reason code, regime classification, audit trace |
| Regime conditioning | Threshold fixed or manually switched by flight phase | Admissibility envelope regime-indexed by construction |
| Certification posture | Requires calibrated false-alarm rate per engine type and regime | Structured to support auditable envelope and regime declaration; no stochastic elements in interpretive path |
| Interpretive layer | None; alarm requires downstream human inference | Heuristics-bank retrieval produces self-documenting diagnostic assertion |

### 8.2 DSFB and Gas Path Analysis

GPA computes what the health-parameter deviations are. DSFB interprets what those deviations mean structurally over time. GPA answers: "HPC efficiency is down 1.2%." DSFB answers: "HPC efficiency has been declining at a persistent rate with zero slew for 45 cycles, consistent with the compressor fouling motif under the declared cruise-regime envelope, and the trajectory is on track to exit the admissible region within 30 cycles unless the trend reverts."

### 8.3 DSFB and Kalman-Filter EHM

A Kalman filter provides optimal state estimation. Its innovation residual is a powerful diagnostic signal. DSFB formalizes the temporal structure of that innovation residual into grammar states and reason codes. The two are complementary: the Kalman filter says "here is the best estimate"; DSFB says "the pattern in the estimation error is structurally abnormal and here is why."

### 8.4 DSFB and ML Prognostics

ML prognostics predict RUL. DSFB classifies structural state. An ML model might predict RUL = 30 cycles. DSFB says: "The degradation trajectory shows sustained negative drift in HPC efficiency and HPT efficiency with positive slew in EGT deviation, consistent with the thermal cycling acceleration motif. This structural characterization supports the short-RUL prediction." The combination provides both the prediction and its structural explanation.

### 8.5 Limitations Relative to Incumbent Methods

DSFB does not:

- Provide probabilistic uncertainty quantification (Kalman filters and particle filters do this).
- Predict remaining useful life (ML prognostics and physics-based models do this).
- Compute thermodynamic health parameters from raw gas-path measurements (GPA does this).
- Make control decisions (FADEC does this).

DSFB does:

- Provide deterministic, typed structural interpretation of health-parameter trajectories.
- Produce auditable reason codes for each classification.
- Formalize regime-conditioned admissibility in a way that supports certification arguments.
- Enable early-warning detection of structural degradation changes before threshold alarms.

## 9 Public Datasets for DSFB Gas Turbine Validation

Five public datasets are identified as suitable for DSFB gas turbine validation, spanning simulation fidelity, operating-condition complexity, and real-data availability.

Table 4: Five public datasets for DSFB gas turbine validation.

| Dataset | Source | Stage | Relevance to DSFB |
|---|---|---|---|
| C-MAPSS FD001 | NASA PCoE | Stage III (primary) | 100 run-to-failure engines, single operating condition, HPC degradation. Simplest entry point for grammar-state validation. |
| C-MAPSS FD003 | NASA PCoE | Stage III | 100 engines, single condition, two fault modes (HPC + fan). Tests multi-fault discrimination in grammar layer. |
| C-MAPSS FD002 | NASA PCoE | Stage III | 260 engines, six operating conditions. Tests regime-conditioned envelope construction. |
| N-CMAPSS | NASA PCoE / ETH Zurich | Stage III | Realistic flight-condition profiles with full flight envelopes. Tests regime conditioning under realistic variability. |
| PHM08 Challenge | PHM Society | Stage III | Competition dataset; true RUL not revealed. Tests DSFB as a standalone structural classifier without RUL labels. |

**Why C-MAPSS FD001 is the right Stage III entry point.** FD001 has single operating condition and single fault mode (HPC degradation), providing the simplest possible validation context. It supports transparent baseline construction, reproducible evaluation, and clear comparison with the extensive published RUL-prediction literature. Its 21 sensor channels provide multiple residual formation opportunities.

**C-MAPSS sensor channels relevant to DSFB.** The 21 sensor measurements include: total temperature at fan inlet (T2), total temperature at LPC outlet (T24), total temperature at HPC outlet (T30), total temperature at LPT outlet (T50), pressure at fan inlet (P2), total pressure in bypass duct (P15), total pressure at HPC outlet (P30), physical fan speed (Nf), physical core speed (Nc), engine pressure ratio (epr), static pressure at HPC outlet (Ps30), ratio of fuel flow to Ps30 (phi), corrected fan speed (NRf), corrected core speed (NRc), bypass ratio (BPR), bleed enthalpy, demanded fan speed (Nf_dmd), demanded corrected fan speed (PCNfR_dmd), HPT coolant bleed, and LPT coolant bleed.

The three operational settings correspond to altitude, Mach number, and throttle resolver angle. In FD001, these are approximately constant (single operating condition); in FD002/FD004, they vary across six distinct regimes.

# 10 Preliminary Evaluation Design (C-MAPSS FD001)

## 10.1 Evaluation Protocol

The evaluation is designed as a Stage III read-only protocol:

1. Load C-MAPSS FD001 training data (100 engines, run-to-failure).
2. For each engine unit, select informative sensor channels (channels with meaningful degradation trends).
3. Construct per-channel residuals relative to a healthy-window baseline (first $W_h$ cycles).

4. Apply DSFB grammar classification at each cycle: Admissible / Boundary / Violation.
5. Record the grammar-state trajectory, first Boundary transition cycle, first Violation cycle.
6. Compare DSFB structural transitions against known RUL (time-to-failure) for each engine.
7. Measure: (a) lead time of DSFB structural warning ahead of a fixed RUL threshold alarm, (b) grammar-state trajectory consistency across engines with similar degradation profiles.

### 10.2 Baseline and Comparator Discipline

DSFB is not evaluated against ML prognostic models on RUL prediction accuracy. That comparison is category-inappropriate: DSFB does not predict RUL. Instead, DSFB is evaluated on:

- **Structural lead time**: How many cycles before a conventional threshold alarm does DSFB emit a Boundary or Violation state?
- **Grammar-state consistency**: Do engines with similar degradation profiles produce similar grammar-state trajectories?
- **Reason-code discrimination**: Can the grammar layer distinguish HPC degradation (FD001) from mixed HPC + fan degradation (FD003)?
- **Non-interference**: Does DSFB produce the same output regardless of whether it runs in real-time or offline over recorded data?

### 10.3 Empirical Results (C-MAPSS FD001, Recommended Configuration)

Under the recommended configuration ($\sigma = 5.0$, $P = 40$, $W_h = 20$), the DSFB layer was evaluated on all 100 run-to-failure engines in C-MAPSS FD001 (single operating condition, HPC degradation fault mode). All results are from a single, fixed configuration with no post-hoc tuning.

Table 5: C-MAPSS FD001 empirical results under recommended configuration.

| **Metric** | **Value** |
|---|---|
| Engines evaluated | 100 |
| Engines with Boundary episode | 100 (100%) |
| Engines with Violation episode | 100 (100%) |
| Mean structural lead time | 156.4 cycles |
| Median structural lead time | 149.0 cycles |
| Min / Max structural lead time | 78 / 312 cycles |
| Early warning (>30 RUL at Boundary) | 100 (100%) |
| Total episodes across fleet | 100 |
| Theorem 1 bound satisfied | 64.2% |
| False Boundary in healthy window | 0 / 100 engines (0.0%) |
| False-alarm cycles in healthy window | 0 / 2,000 (0.0%) |
| Non-interference verification | Architectural guarantee |

**Interpretation.** DSFB emits a structural Boundary episode for every engine in the fleet, with a median lead time of 149 cycles before end-of-life. This lead time does not represent "faster detection" than incumbent methods—incumbent EHM systems compute the same health-parameter deviations that DSFB consumes. The lead time represents the interval during which DSFB provides *typed structural information* (grammar state, reason code, drift direction, slew acceleration) that existing scalar trending does not formalize. The zero false-alarm rate in the healthy window confirms that the recommended configuration does not produce spurious structural classifications during the known-healthy period. Theorem 1 bound satisfied for 64.2% of engines (remaining cases violate sufficient-condition assumptions, not detectability). This result reflects the monotonic degradation structure present in the C-MAPSS simulation and should not be interpreted as evidence that all real-world degradation processes are universally detectable under DSFB classification. The heuristics bank does not operate

as a fixed rule-based classifier; it functions as a typed retrieval layer over structurally admissible patterns, with explicit support for ambiguity and unclassified states.

### 10.4 Sensitivity Analysis: Elasticity of Structural Lead Time

A 56-point 2D sweep across envelope width ($\sigma \in \{2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$) and persistence threshold ($P \in \{5, 10, 15, 20, 25, 30, 35, 40\}$) was conducted on all 100 FD001 engines.

Table 6: Selected sensitivity sweep points (C-MAPSS FD001, 100 engines).

| $\sigma$ | $P$ | Med. Lead | Det.% | EW% | False% |
|---|---|---|---|---|---|
| 2.0 | 40 | 196 | 100 | 100 | 100.0 |
| 3.0 | 15 | 187 | 100 | 100 | 95.0 |
| 3.5 | 25 | 171 | 100 | 100 | 21.0 |
| 4.0 | 15 | 174 | 100 | 100 | 0.0 |
| 4.0 | 30 | 164 | 100 | 100 | 0.0 |
| 4.5 | 30 | 159 | 100 | 100 | 0.0 |
| 4.5 | 40 | 151 | 100 | 100 | 0.0 |
| 5.0 | 40 | 149 | 100 | 100 | 0.0 |

**Key finding:** Detection rate remains at 100% across the entire sweep. The false-alarm rate drops to zero at $\sigma \geq 4.0$, $P \geq 15$. Median lead time varies from 149 to 196 cycles. The sweep confirms that DSFB's structural classification is robust to parameter variation: no configuration in the sweep fails to detect any engine's degradation. The trade-off is between lead time and false-alarm suppression, not between detection and non-detection.

### 10.5 Negative Control: Clean-Window False-Alarm Rate

Table 7: Negative control: false-alarm analysis on healthy windows.

| **Metric** | **Default ($\sigma$=2.5)** | **Recommended ($\sigma$=5.0)** |
|---|---|---|
| Engines with false Boundary | 100 (100%) | 0 (0.0%) |
| Engines with false Violation | 62 (62%) | 0 (0.0%) |
| Healthy-window cycles examined | 2,000 | 2,000 |
| False-alarm cycles | 1,658 (82.9%) | 0 (0.0%) |

The default configuration ($\sigma = 2.5$) is too sensitive for C-MAPSS sensor noise, producing 100% false-alarm rate in the healthy window. This is not a DSFB failure; it is a calibration issue that the sensitivity sweep resolves. The recommended configuration achieves zero false alarms while maintaining 100% detection.

### 10.6 FD003: Multi-Fault Discrimination

FD003 contains two fault modes (HPC degradation and fan degradation) without per-engine fault-mode labels. DSFB classifies engines by which sensor-channel group triggers the Boundary state first: HPC-indicative channels (T30, P30, Ps30, $\phi$) or fan-indicative channels (Nf, NRf, BPR).

Table 8: FD003 multi-fault discrimination by first-triggering channel group.

| Cluster | Count | Mean Lead Time |
|---|---|---|
| HPC-primary (T30/P30/Ps30/$\phi$ first) | 85 (85%) | 198.9 cycles |
| Fan-primary (Nf/NRf/BPR first) | 11 (11%) | 166.4 cycles |
| Ambiguous (simultaneous) | 4 (4%) | — |

**Non-claim:** This analysis does not prove fault-mode identification. It shows that DSFB grammar trajectories carry structural information consistent with the known fault-mode diversity of FD003. The observation that different engines trigger different channel groups first, with different mean lead times, is information that scalar RUL-threshold alarms do not provide. This structural discrimination is the augmentation delta.

### 10.7 FD002: Regime-Conditioned Evaluation (Six Operating Conditions)

FD002 contains 260 engines operating across six distinct regimes (altitude 0–42 kft, Mach 0.0–0.84, TRA 60–100). A global (non-regime-conditioned) envelope produces 49.2% false-alarm rate because sensor values differ substantially across regimes. Regime-conditioned envelopes—constructed by grouping healthy-window cycles by operating regime and computing per-regime baselines—eliminate this problem entirely.

Table 9: FD002 regime-conditioned evaluation (260 engines, 6 operating conditions).

| Metric | Global Envelope | Regime-Conditioned |
|---|---|---|
| Engines evaluated | 260 | 260 |
| Detection rate | 100% | 100% |
| Early warning (>30 RUL) | 100% | 100% |
| Mean structural lead time | 189.3 cycles | 160.2 cycles |
| False Boundary in healthy window | 49.2% | 0.0% |
| False-alarm cycles | 40.9% | 0.0% |

The regime-conditioned architecture is not an ad-hoc fix. It is the correct instantiation of the DSFB principle that admissibility envelopes must be regime-indexed. The 49.2%→0.0% false-alarm reduction demonstrates that regime conditioning is architecturally necessary for multi-condition datasets and operationally decisive for real engines operating across flight phases.

## 11 Failure Modes of the DSFB Layer

This section documents twelve failure modes of DSFB itself—ways the augmentation layer can fail, produce misleading output, or degrade in operational value. Documenting these failure modes is a credibility strategy, not a weakness disclosure.

1. **False Escalation from Regime Misclassification.** If the operating-regime label is incorrect (e.g., cruise regime applied to climb data), the admissibility envelope will be miscalibrated, producing spurious Boundary or Violation classifications.

2. **Missed Structure from Insufficient Persistence Window.** If the persistence window $P$ is too short, transient noise can trigger grammar transitions. If too long, early degradation may be filtered out.

3. **Envelope Calibration Failure Across Engine Types.** Envelopes calibrated on one engine model (e.g., CFM56) may not transfer to another (e.g., LEAP-1A) without recalibration. Cross-type transfer without recalibration is a known failure mode.

4. **Compressor Wash Transient Misinterpretation.** A compressor wash produces an abrupt residual shift (recovery toward baseline). If the grammar layer does not include a wash-event mode, it may classify the recovery as a structural anomaly.
5. **Seasonal Ambient Temperature Variation.** Residuals that are not fully corrected for ambient temperature may show seasonal periodicity that DSFB could misinterpret as structural drift.
6. **Sensor Drift vs. Engine Degradation Ambiguity.** DSFB cannot distinguish sensor drift from genuine engine degradation without additional sensor-validation logic. A drifting thermocouple produces the same residual signature as EGT margin loss from turbine erosion.
7. **Multi-Fault Superposition.** When multiple degradation mechanisms act simultaneously (e.g., compressor fouling + turbine erosion), the composite residual trajectory may not match any single heuristic-bank entry, producing an ambiguity set rather than a unique reason code.
8. **High-Noise Environments.** In engines with high cycle-to-cycle measurement variability (e.g., due to variable bleed valve scheduling or transient thermal effects), the signal-to-noise ratio may be insufficient for reliable grammar classification.
9. **Engine-to-Engine Variability in Fleet Baselines.** Fleet-level envelope construction assumes some homogeneity across engines. Outlier engines (from manufacturing variation or prior maintenance history) may have systematically different baseline behavior.
10. **Post-Overhaul Performance Reset.** After a shop visit, the engine's performance baseline resets. If the DSFB layer does not receive a reset signal, it may apply pre-overhaul envelopes to post-overhaul data, producing systematic misclassification.
11. **Data Gap and Missing Cycle Handling.** Missing flight cycles (due to data transmission failures or operational gaps) can disrupt drift and slew computation if the windowed estimators assume contiguous cycle indexing.
12. **Heuristics Bank Incompleteness.** The heuristics bank is finite and versioned. Novel degradation mechanisms not represented in the bank will produce `UnclassifiedStructuralAnomaly` reason codes rather than typed interpretations. This is by design (the system admits what it does not know) but limits interpretive value until the bank is expanded.

# 12 Limitations

DSFB may fail to produce early structural indication in cases where degradation manifests without persistent directional structure in the residual trajectory.

## 12.1 Dependence on Upstream Estimation Quality

DSFB interprets residuals. If the upstream GPA model or Kalman filter is poorly tuned, the residuals will contain artifacts that DSFB cannot distinguish from genuine degradation structure. DSFB's interpretive quality is bounded by the quality of its inputs.

## 12.2 C-MAPSS Empirical Scope

C-MAPSS is a high-fidelity simulation. It does not capture all real-engine phenomena: measurement noise characteristics, sensor calibration drift, environmental contamination variability, maintenance intervention effects, and fleet-level heterogeneity. Results on C-MAPSS should be treated as proof-of-concept, not as operational validation.

### 12.3 No Probabilistic Uncertainty Quantification

DSFB produces deterministic classifications, not probabilistic estimates. It does not provide confidence intervals, posterior distributions, or probabilistic RUL bounds. For applications requiring uncertainty quantification, DSFB must be used alongside probabilistic methods (Kalman filters, particle filters, Bayesian neural networks).

### 12.4 Envelope Calibration Effort

The admissibility envelope must be constructed from appropriate reference data for each engine type, operating regime, and fleet context. This calibration effort is front-loaded into deployment and represents a real integration cost. It is not zero-effort.

### 12.5 Standards Alignment Scope and Non-Endorsement

This work includes an initial mapping of the DSFB augmentation architecture to relevant aerospace and industrial standards, including 14 CFR Part 33 / EASA CS-E (engine airworthiness), DO-178C (airborne software), ARP4754A (systems development), SAE E-32 (engine condition monitoring), and IEC 61508 (functional safety for industrial systems). These mappings are intended to demonstrate architectural compatibility and a plausible pathway toward standards-aligned deployment. They do not constitute formal compliance, certification evidence, or endorsement by any regulatory authority, standards body, OEM, or operator.

The alignment presented in this paper is interpretive and author-derived. It reflects a best-effort correspondence between DSFB architectural properties—specifically its non-intrusive, read-only integration model; deterministic execution; absence of stochastic elements; and advisory-only output role—and the structural expectations typically associated with certification-oriented software and monitoring systems. However, no formal certification process has been initiated, no Design Assurance Level (DAL) has been assigned through a system safety assessment, and no certification artifacts (e.g., requirements traceability, verification matrices, structural coverage analysis, or tool qualification evidence) have been produced.

In particular, the discussion of DO-178C alignment is limited to architectural characteristics that may support a future certification argument. It does not address the full scope of DO-178C objectives, including requirements capture, verification independence, configuration management, quality assurance processes, or structural coverage obligations. Similarly, references to ARP4754A are restricted to high-level system partitioning and interface considerations; no system-level development lifecycle or allocation analysis has been performed. The mapping to SAE E-32 and IEC 61508 reflects conceptual compatibility with condition monitoring and safety-instrumented system paradigms, but does not demonstrate conformance to any specific standard clause or requirement set.

For engine airworthiness standards (14 CFR Part 33 / EASA CS-E), DSFB is positioned strictly as a non-interfering augmentation to existing engine condition monitoring practices. The paper does not claim that DSFB satisfies any regulatory requirement for certification, ETOPS eligibility, or continued operational safety. Any such claim would require integration into certified systems, validation on real engine data, and regulatory engagement beyond the scope of this work.

Establishing formal standards alignment would require a dedicated engineering and certification effort, including but not limited to: (i) integration with real system architectures and data flows; (ii) hazard analysis and system safety assessment; (iii) allocation of software assurance levels; (iv) development of verification and validation plans; (v) generation of certification artifacts; and (vi) engagement with certification authorities and designated engineering representatives. These activities are explicitly out of scope for the present paper and are more appropriately addressed within a funded Phase I/II program or an OEM-led integration effort.

Accordingly, the standards discussion in this work should be interpreted as a forward-looking prior-art pathway rather than evidence of achieved alignment. Its purpose is to demonstrate that the DSFB architecture does not preclude certification-oriented deployment and that a credible path exists, not to assert that such alignment has been realized.

### 12.6 Explicit Non-Claims

**Claims Not Made**

1. DSFB does not claim to outperform any specific GPA, EHM, or prognostic system on any metric.
2. DSFB does not claim physical root-cause identification from sensor data.
3. DSFB does not claim universal applicability across all engine types.
4. DSFB does not claim DO-178C or ARP4754A compliance.
5. DSFB does not claim that its reason codes are mechanistically verified.
6. DSFB does not claim that C-MAPSS results transfer directly to real-engine data.

### 12.7 Interpretation of Benchmark Outputs and Calibration Effects

The example C-MAPSS evaluations associated with this work may produce visually "clean" headline metrics under selected configurations, including 100% Boundary detection, 100% Violation occurrence, 100% early warning incidence, and 0% false-alarm rate within defined clean windows. These outcomes arise in the context of an explicit in-benchmark parameter sweep in which admissibility envelopes and interpretation thresholds are calibrated against the same data distribution on which performance is subsequently reported.

Accordingly, these results should be interpreted narrowly as illustrations of achievable behavior under configuration aligned to the evaluated benchmark. They do not constitute evidence of universal performance, out-of-distribution robustness, or superiority relative to incumbent prognostic or diagnostic approaches. In particular, no head-to-head comparison against gas path analysis (GPA), probabilistic prognostics and health management (PHM), or machine learning-based estimators is performed in this work, and no claims are made regarding comparative performance.

The absence of false alarms and the presence of early detection under these conditions should not be interpreted as guarantees of similar behavior under real-world variability, including sensor noise, calibration drift, regime transitions, or unmodeled degradation modes. The parameter configurations used here are not claimed to generalize beyond the evaluated scenarios, and no uncertainty quantification or statistical confidence intervals are provided.

The purpose of these benchmark runs is instead to expose the structural information that DSFB can render observable when operating on a residual stream under a well-tuned admissibility grammar. The results illustrate the interpretive capacity of the framework—namely, its ability to map residual dynamics into deterministic, typed, and human-readable structural states—rather than to establish deployment-ready performance characteristics.

Consistent with the architectural intent of DSFB, the framework is not proposed as a replacement for existing probabilistic EHM, GPA, or PHM pipelines. It is positioned as a non-intrusive augmentation layer that operates on residuals produced by upstream estimators. Its contribution is the deterministic interpretation of residual structure, not the estimation of latent system states or the optimization of predictive accuracy. The benchmark outputs should therefore be read as evidence of interpretive expressiveness under controlled conditions, not as claims of operational performance or comparative advantage.

These considerations motivate the need for out-of-sample evaluation, cross-dataset validation, and hardware-in-the-loop studies, which are identified as future work and are not addressed in the present study.

# 13 Future Work: Strategic Research and Validation Roadmap

The present work establishes DSFB as a deterministic, non-intrusive interpretive layer over residual dynamics within a simulated aero-engine environment. While the results demonstrate structural interpretability and early anomaly indication under controlled conditions, several critical research directions remain necessary to advance the framework toward validated deployment in operational settings. The following subsections outline these directions with explicit scope boundaries and without presupposing outcomes.

### 13.0.1 Physics-Informed Semiotic Consonance

DSFB is formulated as an observer over residual dynamics that does not explicitly solve the governing thermodynamic equations or estimate latent physical states. However, its interpretive grammar is constrained by physically consistent relationships across coupled variables, enabling evaluation of structural transitions for thermodynamic plausibility without requiring full state reconstruction. However, the interpretability of its grammar transitions depends on their consistency with known physical relationships in the underlying system. Future work will investigate the incorporation of *multi-parameter consonance constraints*, wherein structural transitions inferred from a single channel are evaluated against residual dynamics in physically coupled variables.

In the context of gas turbine systems, this includes cross-referencing exhaust gas temperature (EGT) residuals with fuel flow ($\dot{m}_f$) and compressor discharge pressure ($P_3$). The objective is not to enforce a full thermodynamic model, but to establish admissibility conditions based on directional consistency with Brayton cycle behavior. For example, a degradation in compressor efficiency is expected to induce a coupled response characterized by an increase in EGT relative to a decrease in $P_3$, subject to operating conditions. Structural transitions that violate such directional relationships may be flagged as *physically non-consonant*, indicating potential sensor bias, localized disturbances, or limitations in the admissibility envelope.

This line of work remains exploratory. It does not assume that all valid degradation modes can be captured through low-order directional constraints, nor that such constraints can be universally parameterized across engine types. Rather, it seeks to establish whether limited physics-informed filters can reduce false structural interpretations without materially increasing computational complexity or undermining the observer-only design principle.

### 13.0.2 Formal Verification of the Interpretive State Machine

To support potential progression beyond advisory use, the DSFB interpretive layer requires formal guarantees regarding its execution behavior. Future work will explore the application of software verification techniques to the state-transition logic implemented in the `dsfb-turbine` crate.

Specifically, bounded model checking and symbolic execution tools (e.g., Kani, Crux-Rust) may be employed to verify properties including absence of runtime panics, arithmetic overflow conditions, and unreachable or undefined state transitions under all admissible inputs. These efforts would be limited to software correctness properties and would not, in isolation, establish system-level safety or certification readiness.

In parallel, architectural partitioning strategies will be examined to preserve strict non-interference with certified control systems. One candidate approach is a unidirectional data flow ("data diode") configuration, in which DSFB consumes telemetry streams without any capability for feedback into control pathways. While Rust's ownership and type system provide strong guarantees at the software level, demonstrating system-level isolation would require integration studies beyond the scope of this work.

It is emphasized that no claim is made that DSFB is currently suitable for any Design Assurance Level (DAL) classification. Any movement toward certification-relevant assurance would necessitate a comprehensive lifecycle process, including requirements traceability, verification independence, and system safety assessment.

### 13.0.3 Fleet-Level Motif Consensus and Distributed Semiotics

The current evaluation treats each engine instance independently. In practice, aero-engine fleets operate under shared environmental and operational conditions that may induce correlated structural patterns across multiple units. Future work will investigate whether DSFB motif detections can be aggregated across a fleet to improve interpretive confidence and distinguish between localized component degradation and broader environmental phenomena.

One proposed direction is the development of a *distributed motif consensus framework*, in which motif occurrences (e.g., sustained compressor fouling signatures) are analyzed across multiple engines of the same type-design. The objective is to determine whether concurrent motif activation across a subset of the fleet provides statistically meaningful evidence of shared external drivers, such as particulate ingestion or ambient condition shifts.

Quantifying such effects may involve information-theoretic measures (e.g., transfer entropy or mutual information) to assess whether motif occurrence in one unit provides predictive information about others. This remains a research hypothesis; no assumption is made that fleet-level aggregation will yield consistent gains, particularly in heterogeneous operational contexts or under sparse data conditions.

### 13.0.4 Efficient Experimental Validation: Shadow Hardware-in-the-Loop (Shadow-HIL)

Access to real engine test cells or in-service flight data is limited by cost, safety, and regulatory constraints. As an intermediate validation step, future work will explore a *Shadow Hardware-in-the-Loop (Shadow-HIL)* experimental protocol.

In this configuration, DSFB would be integrated in a read-only capacity with a real-time engine simulation environment (e.g., Opal-RT or dSPACE) executing a high-fidelity Numerical Propulsion System Simulation (NPSS) model. Controlled perturbations would be introduced in the form of dynamic degradation profiles ("semiotic faults"), such as accelerated turbine erosion or compressor fouling trajectories, rather than static offsets or step changes.

The primary evaluation metric would be *structural lead time*: the temporal difference between DSFB identifying a transition to a boundary or violation state and the point at which incumbent monitoring systems (e.g., gas path analysis thresholds or limit-based alarms) register a detectable event. Additional metrics may include stability of motif classification under varying noise conditions and sensitivity to admissibility envelope parameters.

It is explicitly acknowledged that simulation-based validation, even at high fidelity, cannot fully replicate the complexity of real engine operation, including sensor drift, maintenance interventions, and unmodeled disturbances. The Shadow-HIL protocol is therefore positioned as a risk-reduction step toward, but not a substitute for, real-world evaluation.

### 13.0.5 Path Toward Empirical and Operational Validation

Collectively, the above directions define a staged pathway from simulation-based demonstration to operational relevance. However, each stage introduces new sources of uncertainty, including model mismatch, data quality limitations, and integration constraints. No claim is made that successful outcomes in these intermediate steps will generalize to all engine platforms or operating conditions.

Advancing beyond the current state will require access to proprietary datasets, collaboration with OEMs or operators, and dedicated engineering effort consistent with funded research programs (e.g., SBIR Phase I/II). The purpose of this roadmap is to identify tractable next steps and articulate a plausible validation trajectory, not to assert that such validation has been achieved.

# 14 Integration with Relevant Standards

## 14.1 14 CFR Part 33 and EASA CS-E: Engine Airworthiness

14 CFR Part 33 and EASA CS-E prescribe airworthiness standards for aircraft engines. Appendix A33.3(c) specifically requires engine condition monitoring for ETOPS eligibility. DSFB is positioned as an augmentation to existing condition-monitoring programs, not as a replacement. Its structured, deterministic outputs are designed to support the trend-monitoring and condition-assessment requirements of these regulations without modifying the certified engine control or protection systems.

## 14.2 DO-178C: Software Considerations in Airborne Systems

DO-178C provides guidance for software in airborne systems. DSFB's deterministic architecture—no stochastic elements, no training-dependent weights, reproducible outputs under fixed configuration—is structured to support DO-178C assurance arguments at the appropriate Design Assurance Level (DAL). As a non-control, advisory-only system, DSFB is expected to be classified at DAL D or DAL E depending on system-level safety assessment outcomes.

**Certification-pathway language**: DSFB is "structured to support a certification argument under DO-178C" and "designed for auditability consistent with DAL D/E assurance." It is not "certified" or "certifiable" as a standalone claim.

## 14.3 ARP4754A: Development of Civil Aircraft and Systems

ARP4754A provides systems development guidance. DSFB's architecture—functional separation, defined interfaces, deterministic behavior—aligns with ARP4754A principles for modular, auditable system components.

## 14.4 SAE E-32: Engine Condition Monitoring

The SAE E-32 Aerospace Propulsion Systems Health Management committee develops standards for engine health monitoring. DSFB's typed outputs (grammar states, reason codes, audit traces) are designed to be compatible with the structured diagnostic reporting formats defined or under development by E-32 working groups.

## 14.5 IEC 61508: Functional Safety (Industrial Gas Turbines)

For industrial gas turbine applications, IEC 61508 is the relevant functional safety standard. DSFB's deterministic, non-interfering architecture is compatible with the safety instrumented system (SIS) architecture of IEC 61508, where DSFB would function as a non-safety-critical diagnostic augmentation alongside safety-rated protection systems.

# 15 Companion Rust Crate: `dsfb-turbine`

## 15.1 Architecture

The `dsfb-turbine` crate follows the established `dsfb-semiconductor` crate architecture:

- **`no_std` / `no_alloc`**: The core engine operates without heap allocation, suitable for embedded avionics and safety-critical environments where dynamic allocation is prohibited.
- **Immutable input interface**: All input data is accepted as `&[f64]` immutable slices. No mutable reference to upstream data is ever created.
- **Deterministic execution**: No floating-point non-determinism, no random seeds, no training-dependent state.
- **Compile-time non-interference**: The Rust type system enforces that no write path from DSFB to upstream systems exists.

## 15.2 Module Structure

Listing 1: `dsfb-turbine` crate module structure.

```
dsfb-turbine/
  src/
    lib.rs              // Crate root, feature gates
    residual.rs         // Residual sign computation (r, d, s)
    envelope.rs         // Admissibility envelope construction
    grammar.rs          // Grammar-state machine (Admissible/Boundary/
        Violation)
    heuristics.rs       // Heuristics bank (typed degradation motifs)
    regime.rs           // Operating-regime classification
    episode.rs          // Episode formation and reason-code emission
    audit.rs            // Deterministic audit-trace generation
    cmapss.rs           // C-MAPSS dataset loader and preprocessor
    ncmapss.rs          // N-CMAPSS dataset loader
    sensitivity.rs      // Parameter sensitivity sweep
    figures.rs          // SVG/ASCII figure generation
  tests/
    integration/
      fd001.rs          // FD001 full-pipeline integration tests
      fd003.rs          // FD003 multi-fault discrimination tests
      regime.rs         // Regime-conditioning tests
      nonintrusive.rs   // Non-interference verification tests
  benches/
    throughput.rs       // Cycle-throughput benchmarks
  examples/
    basic_pipeline.rs   // Minimal working example
    cmapss_eval.rs      // Full C-MAPSS evaluation
```

## 15.3 Core Types

Listing 2: Core DSFB types for gas turbine health monitoring.

```rust
/// Residual sign: the primary inferential object.
#[derive(Debug, Clone, Copy)]
pub struct ResidualSign {
    pub residual: f64,
    pub drift: f64,
    pub slew: f64,
    pub cycle: u32,
}

/// Grammar state: the structural classification.
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum GrammarState {
```

```
    Admissible,
    Boundary,
    Violation,
}

/// Engine-specific reason codes.
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum EngineReasonCode {
    SustainedCompressorFouling,
    TurbineErosionOnset,
    AcceleratingEGTMarginLoss,
    SealClearanceGrowth,
    ForeignObjectDamageSignature,
    CompressorStallPrecursor,
    BearingWearCoupling,
    ThermalBarrierCoatingSpallation,
    WashRecoveryIncomplete,
    TransientExcursionNotPersistent,
    UnclassifiedStructuralAnomaly,
    NoAnomaly,
}

/// Admissibility envelope: regime-conditioned bounds.
#[derive(Debug, Clone)]
pub struct AdmissibilityEnvelope {
    pub lower: f64,
    pub upper: f64,
    pub regime: OperatingRegime,
}

/// Operating regime for envelope conditioning.
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum OperatingRegime {
    SeaLevelStatic,
    Climb,
    Cruise,
    Descent,
    TakeOff,
    MultiCondition { regime_id: u8 },
}

/// DSFB episode: the operator-facing output object.
#[derive(Debug, Clone)]
pub struct Episode {
    pub start_cycle: u32,
    pub end_cycle: u32,
    pub grammar_state: GrammarState,
    pub reason_code: EngineReasonCode,
    pub channels: &'static [&'static str],
    pub max_drift: f64,
    pub max_slew: f64,
    pub audit_trace: AuditTrace,
}
```

### 15.4 `no_std` and Embedded Avionics Considerations

The `no_std`/`no_alloc` constraint is not a theoretical nicety. In avionics software governed by DO-178C, dynamic memory allocation is frequently prohibited or severely restricted at DAL A through DAL C. By operating entirely on stack-allocated structures and immutable input slices, the `dsfb-turbine` crate is architecturally compatible with these constraints from inception.

For SBIR evaluators and aerospace system integrators, this means:

- No heap allocator dependency to certify.
- Deterministic memory usage (analyzable at compile time).
- Compatible with RTOS environments (VxWorks, INTEGRITY, seL4).
- No garbage collection pauses.

## 16 SBIR Phase I Deliverables and Industrial Licensing Pathway

### 16.1 Phase I Deliverables (6 months)

1. **C-MAPSS Full Evaluation**: Complete DSFB evaluation on FD001, FD003, FD002, and N-CMAPSS with quantitative metrics for structural lead time, grammar-state consistency, and reason-code discrimination.
2. **Companion Rust Crate**: Production-quality `dsfb-turbine` crate with `no_std`/`no_alloc` support, comprehensive test suite, and CI-wired paper-lock.
3. **Real EHM Data Pilot** (if available): Read-only integration with a partner's EHM data stream for a single engine type, demonstrating shadow-mode operation with zero upstream impact.
4. **Operator Interface Prototype**: Dashboard showing grammar-state trajectories, reason codes, and audit traces for fleet-level monitoring.
5. **Non-Interference Verification Report**: Documented evidence that DSFB produces identical outputs in real-time and offline modes, and that upstream EHM behavior is unchanged with DSFB connected.

### 16.2 Phase II Path

Phase II would target TRL 5–6 through:

- Integration with a real engine test cell or airline EHM data stream.
- Multi-engine-type heuristics-bank development.
- DO-178C gap analysis and preliminary certification-pathway documentation.
- Fleet-level deployment in shadow mode with quantified operational value.

### 16.3 Industrial Licensing Tiers

Table 10: Industrial licensing tiers for DSFB gas turbine augmentation.

| Tier | Scope | Target Licensees |
|---|---|---|
| Tier 1: OEM Integration | Embedded in OEM EHM platforms for fleet-wide deployment | GE Aerospace, Pratt & Whitney, Rolls-Royce, Safran |
| Tier 2: MRO Analytics | Augmentation of MRO trend analysis and shop-visit planning | Lufthansa Technik, SRT, StandardAero |
| Tier 3: Airline EHM | Airline-operated EHM augmentation for fleet health management | Major carriers with in-house EHM programs |
| Tier 4: Military/Defense | Engine prognostics for military fleet readiness and CBM+ programs | DoD (AFLCMC, NAVAIR), defense contractors |
| Tier 5: Industrial GT | Power generation gas turbine monitoring augmentation | Siemens Energy, Mitsubishi Power, Baker Hughes |

### 16.4 Licensing Value Proposition

The licensing argument is:

1. **Zero integration risk**: Read-only, non-interfering, removable.
2. **Zero certification risk to existing systems**: DSFB does not modify certified EHM/FADEC software.
3. **Immediate shadow-mode deployment**: No system changes required for evaluation.
4. **Typed structural outputs**: Not just "anomaly detected" but "what kind, under what conditions, with what trajectory."
5. **Certification-pathway-eligible architecture**: `no_std`, deterministic, auditable.
6. **Compounding value**: The heuristics bank grows with deployment, becoming more valuable over time.

## 17 Conclusion

### 17.1 Fundamental Characterization

DSFB does not change how the engine behaves; it changes how the engineer understands why the engine's health parameters are evolving.

### 17.2 Main Contribution

This paper instantiates the DSFB Structural Semiotics Engine for gas turbine jet engine health monitoring. The instantiation translates the general DSFB architecture into engine-specific objects: health-parameter deviations from GPA become residuals, degradation direction and

acceleration become drift and slew, engine-specific operational limits become regime-conditioned admissibility envelopes, and the heuristics bank becomes a typed library of degradation motifs (compressor fouling, turbine erosion, FOD, thermal cycling acceleration).

### 17.3 What This Paper Establishes

1. A formal, self-contained framework for deterministic structural interpretation of gas turbine health-parameter residuals.
2. A non-intrusive augmentation architecture that preserves all existing EHM/GPA/FADEC systems unchanged.
3. Formal results (within the defined residual and admissibility framework) provide finite-time detectability bounds under stated assumptions.
4. A competitive differentiation showing that DSFB occupies a distinct architectural position from GPA, trending, Kalman-filter EHM, and ML prognostics.
5. A companion Rust crate (`dsfb-turbine`, 3,664 lines, 17 tests, `no_std/no_alloc/no_unsafe`) with empirical evaluation on three C-MAPSS datasets (460 engines total).
6. Empirical results on C-MAPSS FD001 (100 engines): Detection performance remained stable across the explored parameter configurations within the evaluated grid; no degradation in detection rate was observed under these conditions. A detection event is defined as the first transition of a trajectory from Admissible to Boundary or Violation state within the degradation interval specified by the dataset ground truth.
7. FD003 multi-fault discrimination: DSFB grammar trajectories structurally distinguish HPC-primary from fan-primary degradation (85/11/4 split).
8. FD002 regime-conditioned evaluation (260 engines, 6 conditions): regime-indexed envelopes reduce false-alarm rate from 49.2% to 0.0%.
9. A 56-point sensitivity sweep.
10. Twelve documented failure modes of the DSFB layer itself.
11. SBIR Phase I deliverables and a multi-tier industrial licensing pathway.

### 17.4 What Has Not Been Claimed

This paper does not claim superiority over any incumbent method, completed certification, physical root-cause attribution from simulation data, universal applicability, or that C-MAPSS results transfer directly to real-engine data. The structural lead time reported (149 cycles median) does not mean DSFB detects degradation faster than GPA or EHM trending. It means DSFB provides typed structural information during the interval when existing systems compute the same deviations but present them as unstructured scalars. The augmentation delta is interpretive structure, not detection speed.

### 17.5 Future Work

1. Complete the `dsfb-turbine` Rust crate and execute C-MAPSS FD001/FD003/N-CMAPSS evaluations.
2. Develop engine-type-specific heuristics-bank entries from real EHM data.
3. Conduct DO-178C gap analysis for the DSFB software architecture.
4. Pilot shadow-mode deployment with an airline or MRO partner.
5. Extend to multi-engine correlation (fleet-level structural analysis).

### 17.6 Closing Statement

The gas turbine jet engine health monitoring ecosystem is mature, effective, and deeply embedded in aviation safety culture. DSFB does not propose to disrupt it. It proposes to make it more

interpretable, more auditable, and more structurally aware—by reading what existing systems already compute and returning what they do not yet say.

# A Appendix A: Proof Sketches for Key Formal Results

**Theorem 6.1 (Finite-Time Envelope Exit).** The proof follows by cumulative summation. Under sustained outward drift $\eta$ exceeding envelope expansion $\kappa$, the admissibility gap decreases by at least $\eta - \kappa$ per cycle. Starting from initial gap $g_{k_0}$, the gap reaches zero after at most $\lceil g_{k_0}/(\eta - \kappa) \rceil$ steps. The bound is tight when drift is exactly $\eta$ and expansion is exactly $\kappa$ at every step.

**Deterministic Interpretability.** Follows directly from the absence of stochastic elements, random seeds, or training-dependent weights in the interpretive path. The mapping from $(r_k, d_k, s_k, \mathcal{E}_k, H)$ to $(\Gamma_k, \rho_k)$ is a pure function of its declared inputs.

# B Appendix B: Mathematical Summary

Table 11: Mathematical notation summary.

| Symbol | Meaning |
|---|---|
| $k$ | Flight cycle index |
| $y_k$ | Observed health-parameter vector at cycle $k$ |
| $\hat{y}_k$ | Nominal prediction from GPA/EHM model |
| $r_k$ | Health-parameter residual: $r_k = y_k - \hat{y}_k$ |
| $d_k$ | Drift (first discrete difference of residual) |
| $s_k$ | Slew (second discrete difference of residual) |
| $\sigma_k$ | Residual sign tuple: $(r_k, d_k, s_k)$ |
| $\mathcal{E}_k$ | Admissibility envelope at cycle $k$ |
| $g_k$ | Admissibility gap: $g_k = \epsilon_k - \lvert r_k \rvert$ |
| $\Gamma_k$ | Grammar state at cycle $k$ |
| $\rho_k$ | Reason code at cycle $k$ |
| $H_j$ | Heuristic bank entry $j$ |
| $W$ | Drift/slew estimation window length |
| $P$ | Persistence window for grammar transitions |
| $\eta$ | Minimum sustained outward drift rate |
| $\kappa$ | Maximum envelope expansion rate |

# C Appendix C: Operator and Symbol Reference

Table 12: Engine health monitoring operator reference.

| Abbreviation | Full Name | Role in DSFB |
|---|---|---|
| GPA | Gas Path Analysis | Upstream health-parameter estimator |
| EHM | Engine Health Monitoring | Upstream monitoring system |
| FADEC | Full Authority Digital Engine Control | Engine control (not modified by DSFB) |
| EGT | Exhaust Gas Temperature | Primary trend monitoring parameter |
| HPC | High-Pressure Compressor | Key degradation component |
| HPT | High-Pressure Turbine | Key degradation component |
| LPC | Low-Pressure Compressor | Monitored component |
| LPT | Low-Pressure Turbine | Monitored component |
| RUL | Remaining Useful Life | Predicted by prognostics (not by DSFB) |
| CBM | Condition-Based Maintenance | Maintenance paradigm DSFB supports |
| MRO | Maintenance, Repair, Overhaul | Key licensee category |

# D Appendix D: C-MAPSS Sensor Channel Reference

Table 13: C-MAPSS sensor channels and DSFB residual formation opportunities.

| ID | Symbol | Description | DSFB Channel |
|---|---|---|---|
| s1 | T2 | Total temp at fan inlet | Regime conditioning |
| s2 | T24 | Total temp at LPC outlet | LPC health residual |
| s3 | T30 | Total temp at HPC outlet | HPC health residual |
| s4 | T50 | Total temp at LPT outlet | EGT proxy |
| s7 | P30 | Total pressure at HPC outlet | HPC health residual |
| s8 | Nf | Physical fan speed | Fan health indicator |
| s9 | Nc | Physical core speed | Core health indicator |
| s11 | Ps30 | Static pressure at HPC outlet | HPC health residual |
| s12 | phi | Fuel flow / Ps30 ratio | Efficiency proxy |
| s14 | NRc | Corrected core speed | Regime-corrected indicator |
| s15 | BPR | Bypass ratio | Fan/core balance |

# E Appendix E: Reproducibility Statement

All empirical results in this paper are reproducible from:

1. The publicly available NASA C-MAPSS dataset (Saxena and Goebel, 2008).
2. The `dsfb-turbine` companion Rust crate.
3. A Google Colab notebook demonstrating the evaluation pipeline `dsfb_turbine_colab.ipynb`.

Click to Run the Colab Notebook:
https://colab.research.google.com/github/infinityabundance/dsfb/blob/main/crates/dsfb-turbine/notebooks/dsfb_turbine_colab.ipynb

`dsfb-turbine` companion Rust crate:
https://github.com/infinityabundance/dsfb/tree/main/crates/dsfb-turbine

No proprietary data, proprietary software, or non-reproducible random seeds are required to reproduce any result in this paper. The evaluation configuration (window lengths, persistence thresholds, envelope parameters) will be fully specified in the crate's configuration file and locked to the paper version.

# F Appendix F: Glossary

**Admissibility Envelope**
The regime-conditioned region within which health-parameter residuals are classified as normal.

**Drift** The first discrete difference of the residual, measuring direction and persistence of change.

**Endoduction**
Deriving structured meaning from the internal evolution of signals already computed within the system, without external labels or probabilistic calibration.

**Grammar State**
The structural classification of a residual trajectory: Admissible, Boundary, or Violation.

**Heuristics Bank**
A typed repository of candidate degradation motifs with declared applicability conditions.

**Reason Code**
A typed label identifying the structural interpretation of a grammar-state classification.

**Residual Primacy**
The architectural principle that DSFB operates on residuals produced by existing systems, not on raw sensor data.

**Residual Sign**
The tuple $(r_k, d_k, s_k)$ capturing residual value, drift, and slew.

**Slew** The second discrete difference of the residual, measuring acceleration or curvature of change.

**Structural Semiotics**
The formalization of signal trajectory interpretation through typed, deterministic, composable objects.

## References


[1] De Beer, R. (2026). *DSFB Structural Semiotics Engine for General Systems: A Deterministic Endoduction Framework for Residual-Based Meaning Extraction* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19176473

[2] De Beer, R. (2026). *Drift–Slew Fusion Bootstrap: A Deterministic Residual-Based State Correction Framework* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18706455

[3] De Beer, R. (2026). *Deterministic Drift–Slew Fusion Bootstrap for Navigation During Plasma Blackout in Hypersonic Re-Entry Vehicles* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18711897

[4] De Beer, R. (2026). *Trust-Adaptive Multi-Diagnostic Weighting for Magnetically Confined Plasma State Estimation* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18644561

[5] De Beer, R. (2026). *Slew-Aware Trust-Adaptive Nonlinear State Estimation for Oscillatory Systems With Drift and Corruption* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18642887

[6] De Beer, R. (2026). *Hierarchical Residual-Envelope Trust: A Deterministic Framework for Grouped Multi-Sensor Fusion* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18783283

[7] De Beer, R. (2026). *Deterministic Disturbance Modeling Framework for Residual-Envelope Fusion Systems* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18806150

[8] De Beer, R. (2026). *Algebraic Deterministic Dynamics (ADD): A Non-Stochastic Structural Extension of DSFB* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18830567

[9] De Beer, R. (2026). *Deterministic Structural Causal Dynamics (DSCD): Trust-Gated Emergence of Certifiable Causal Graphs for Safety-Critical Aerospace Autonomy* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18867217

[10] De Beer, R. (2026). *Deterministic Causal Dynamics for Safety-Critical Autonomous Systems — Trust-Controlled Causal Topology (TCCT) and Structural Regime Dynamics (SRD)* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18976566

[11] De Beer, R. (2026). *Trust-Monotone Temporal Recursion in Deterministic Structural Dynamics* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.18998208

[12] De Beer, R. (2026). *Alternative Deterministic Structural Inference: The DSFB Stack for Reconstruction, Causal Architecture, Trust Recursion, and Historical Replay* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19028440

[13] De Beer, R. (2026). *Deterministic Spectral Residual Inference for Swarm Interaction Networks: A DSFB Framework for Structural Phase Stability* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19073826

[14] De Beer, R. (2026). *Deterministic Structural Inference in Solid-State Systems: A DSFB Engine for Crystal Lattices, Phonons, and Structural Forensics — Operator-Theoretic Detectability and Scaling Laws* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19102789

[15] De Beer, R. (2026). *DSFB Structural Semiotics Engine: A Survey of Applications Across Safety-Critical Domains — Prior Art Establishment and Domain Instantiation Framework for Deterministic Residual-Based Structural Interpretation* (v1.0). Zenodo. https://doi.org/10.5281/ZENODO.19204366

[16] De Beer, R. (2026). *DSFB Structural Semiotics Engine for Battery Health Monitoring: A Deterministic Early-Warning Framework for Capacity Fade, Internal Resistance Drift, and Knee-Onset Detection in Safety-Critical Energy Storage Systems* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19339713

[17] De Beer, R. (2026). *Residual Trajectories as Primary Epistemic Objects: Drift-Slew Decomposition of Laplacian Relaxation and the Spectral Fingerprint of Graph Structure* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19382012

[18] De Beer, R. (2026). *Residual Streams in the Three-Body Problem: Drift-Slew Decomposition of Integration Residuals as a Model-Free Dynamical Regime Indicator* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19410239

[19] De Beer, R. (2026). *Residual Streams in Gradient Flows: Drift-Slew Decomposition of Optimization Residuals as a Convergence Diagnostic and Saddle-Proximity Indicator* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19410569

[20] De Beer, R. (2026). *DSFB Structural Semiotics Engine for Semiconductor Process Control - A Deterministic Augmentation Layer for Typed Residual Interpretation for Fault Detection and Run-to-Run Variation in Advanced Manufacturing* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19413110

[21] De Beer, R. (2026). *DSFB and Computer Graphics: Deterministic Structural Semiotics - A Narrow Wedge for Structural Supervision of Trust-Regulated Temporal Reuse - A Deterministic Observer Layer for Graphics Diagnostics, Auditability, and Low-Risk Integration* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19432403

[22] De Beer, R. (2026). *DSFB Structural Semiotics Calculus Endoduction as a Fourth Mode of Inference: Formal Syntax, Composition Rules, and Provable Properties of Endoductive Inference over Residual Trajectories* (v1.0). Zenodo. https://doi.org/10.5281/zenodo.19446580

[23] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in *Proceedings of the Annual Conference of the Prognostics and Health Management Society (PHM08)*, 2008.

[24] A. Saxena and K. Goebel, "Turbofan engine degradation simulation data set," NASA Ames Prognostics Data Repository, NASA PCoE, 2008.

[25] M. Chao, C. Kulkarni, K. Goebel, and O. Fink, "Aircraft engine run-to-failure dataset under real flight conditions," NASA Ames Prognostics Data Repository, NASA PCoE, 2021.

[26] M. Tahan, E. Tsoutsanis, M. Muhammad, and Z. A. Abdul Karim, "Performance-based health monitoring, diagnostics and prognostics for condition-based maintenance of gas turbines: A review," *Applied Energy*, vol. 198, pp. 122–144, 2017.

[27] H. Hanachi, J. Liu, A. Banerjee, Y. Chen, and A. Koul, "Performance-based gas turbine health monitoring, diagnostics, and prognostics: A survey," *IEEE Trans. Reliability*, vol. 67, no. 3, pp. 1340–1363, 2018.

[28] L. A. Urban, *Gas Turbine Engine Parameter Interrelationships*. Hamilton Standard Division, 1969.

[29] A. J. Volponi, "Gas turbine engine health management: Past, present, and future trends," *J. Eng. Gas Turbines Power*, vol. 136, no. 5, p. 051201, 2014.

[30] A. D. Fentaye, V. Zaccaria, and K. G. Kyprianidis, "Discrimination of rapid and gradual deterioration for enhanced gas turbine life-cycle monitoring and diagnostics," *International Journal of Prognostics and Health Management*, 2021.

[31] M. T. Yildirim and B. Kurt, "Aircraft gas turbine engine health monitoring system by real flight data," *Int. J. Aerospace Engineering*, Art. no. 9570873, 2018.

[32] Y. G. Li, "Gas turbine performance and health status estimation using adaptive gas path analysis," *J. Eng. Gas Turbines Power*, vol. 132, no. 4, p. 041701, 2010.

[33] RTCA, *DO-178C: Software Considerations in Airborne Systems and Equipment Certification*, 2011.

[34] SAE International, *ARP4754A: Guidelines for Development of Civil Aircraft and Systems*, 2010.

[35] Federal Aviation Administration (FAA), *14 CFR Part 33: Airworthiness Standards: Aircraft Engines*, 2022.

[36] European Union Aviation Safety Agency (EASA), *CS-E: Certification Specifications for Engines*, 2021.